

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - CYBER SECURITY

Report on Domain Specific Training –on Ethical Hacking

Academic Year 2025–26

The Department of Computer Science and Engineering- Cyber Security at ATME College of Engineering is dedicated to preparing students for successful careers in the rapidly evolving field of Cybersecurity. In addition to providing a strong academic foundation, the department focuses on developing specialized technical competencies through Domain Specific Training (DST). The objective of DST is to bridge the gap between theoretical concepts and industry requirements by providing students with practical, hands-on experience in emerging technologies.

Cybersecurity has become one of the most critical domains in today's digital world, with organizations across every sector facing increasingly sophisticated cyber threats. To address this growing demand, the department has identified **Ethical Hacking** as one of its major career pathways. The structured training enables students to acquire industry-relevant skills, practical knowledge, and professional certifications that significantly enhance their employability and also emphasizes on combining classroom learning with domain-specific training and practical exposure.

Before commencing the technical training, the department organized an awareness program on Domain Specific Training (DST) focusing on Ethical Hacking. The session introduced students to the importance of specialized learning in cybersecurity and highlighted the increasing demand for skilled professionals in the global cybersecurity workforce and application areas of ethical hacking, including banking and financial services, cloud security, web application security, IoT security, government organizations, military infrastructure, healthcare systems, and enterprise cybersecurity.

As part of the Domain Specific Training initiative for the Academic Year 2025–26, the Department of Computer Science and Engineering (Cyber Security) organized an intensive training program on Ethical Hacking for all 5th Semester students in association with HACFY. HacFy is a cybersecurity firm delivering security assessments, penetration testing, cybersecurity training, and security product research and development across critical digital environments. The training was conducted at ATME College of Engineering, Mysuru Campus, from 26 September 2025 to 15 November 2025.

The program consisted of 60 hours of practical and industry-oriented learning covering the complete penetration testing lifecycle. The training emphasized hands-on learning using professional cybersecurity tools and techniques commonly used by penetration testers and security analysts. Students progressed from reconnaissance and vulnerability assessment to exploitation, system hacking, web application security, wireless security, cloud security, vulnerability assessment and penetration testing (VAPT), and professional security report writing.

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Throughout the program, students performed practical exercises in controlled laboratory environments, gaining experience with vulnerability scanning, penetration testing, exploitation techniques, and security assessment methodologies. The project-based approach significantly improved students analytical thinking, problem-solving abilities, teamwork, and technical confidence.

Major Topics Covered

- Reconnaissance and Information Gathering
- Advanced Network Scanning using Nmap
- Vulnerability Assessment and Analysis
- Hands-on Scanning of Vulnerable Virtual Machines (Metasploitable)
- Exploitation Techniques
- System Hacking
- Web Application Hacking
- Vulnerability Assessment and Penetration Testing (VAPT)
- Penetration Testing Methodology
- Wireless Security
- Cloud Security
- Security Report Writing
- Hands-on Practical Penetration Testing Assignment

Conclusion

The Domain Specific Training on Cybersecurity and Ethical Hacking successfully achieved its objective of preparing students for careers in one of the fastest-growing domains in Information Technology, while the 60-hour hands-on training equipped them with practical skills in penetration testing, vulnerability assessment, system security, web application security, wireless security, cloud security, and professional security reporting.

By integrating awareness, structured domain-specific training, practical laboratory exercises, assessments, and project-based learning, the Department of Computer Science and Engineering-Cyber Security has created a comprehensive learning ecosystem that bridges academic education with industry expectations.

This initiative reflects the department's commitment to producing industry-ready cybersecurity professionals who possess practical expertise, strong ethical values, and the ability to contribute effectively to organizational security frameworks. Through continuous collaboration with industry partners and a focus on emerging technologies, ATME College of Engineering continues to empower students with the knowledge and skills required to address the evolving challenges of cybersecurity.

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